

Lecture 14: The Binomial Distribution

Running example: the Milgram experiment

The Binomial Setting

Examples:

Non-example:

The Binomial Distribution

Two common use cases:

Possible values for X :

Probabilities:

Example: You play Monopoly with your siblings a lot — so much so that you have found that you win roughly 1 out of every 20 games played. If you and your siblings will play 20 (independent) games of Monopoly this week, what's the probability that ...

- ... you win at least once?
- ... you win exactly twice?
- ... you win at least twice?
- ... you win between 1 and 5 games?

Example: According to a 2020 Gallup poll, 43% of American voters identify with the Mario Party. Suppose I sample n voters at random and ask for their party identification. If X represents the number of sampled voters who identify with the Mario Party, then X has a _____ distribution.

- If I sample 300 American voters, how many would I expect to identify with the Mario Party?
- If I sample n American voters, how many would I expect to identify with the Mario Party?

Mean and variance for a binomial random variable

Back to Milgram example:

Connection between binomial and normal distributions

Example: In 100 tosses of a fair coin ...

- ... what's the probability of 35 or fewer heads?
- ... what's the probability that the number of heads is within 10% of the expected number?

Example: A college has offered admission to 1200 students this cycle. The number of admitted students who will accept the offer is random; from past experience, roughly 75% of admitted students accept.

- Mean and standard deviation of the number accepting admission?
- Probability that more than 950 students will accept?
- If they decide to admit an additional 100 students off the wait list, what's the probability that more than 950 students in total will accept?